

Digital Farm Management Portal for Implementing Biosecurity Measures in Pig and Poultry Farms

Sprint 1 — Day-by-Day Project Documentation
June 01 – June 13, 2025
Version 1.0

III AIML_A | 2028 Batch | Day WSIE Project Tracker | Planning & Design Sprint

Day	Date	Planned Activity	Expected Deliverable
Day 1	Jun-01	Project Title Finalization	Approved Project Title
Day 2	Jun-02	Requirement Gathering	Problem Statement
Day 3	Jun-03	Objective Definition	Project Objectives
Day 4	Jun-04	User & Module Identification	Module List
Day 5	Jun-05	Use Case Diagram Preparation	CRUD APIs
Day 6	Jun-06	Database Requirement Analysis	Table List
Day 7	Jun-07	ER Diagram Design	ER Diagram
Day 8	Jun-08	Database Schema Creation	SQL Schema
Day 9	Jun-09	UI Wireframe Design	Page Layouts
Day 10	Jun-10	Login & Dashboard UI Design	UI Screens
Day 11	Jun-11	Navigation & Form Design	UI Prototype
Day 12	Jun-12	Design Review	Design Approval
Day 13	Jun-13	Frontend Environment Setup	React Project Setup

1.1 Chosen Project Title

Digital Farm Management Portal for Implementing Biosecurity Measures in Pig and Poultry Farms

1.2 Why This Title Was Chosen

The title was finalized after evaluating multiple alternatives. The phrase 'Digital Farm Management Portal' conveys a comprehensive, technology-driven platform approach. 'Biosecurity Measures' clearly states the primary function — controlling disease spread and contamination. 'Pig and Poultry Farms' anchors the specific agricultural domain, making the scope immediately clear to any evaluator or industry stakeholder.

Three alternative titles were considered and rejected:

Alternative Title	Reason for Rejection
Smart Livestock Monitoring System	Too generic; does not highlight biosecurity or farm management portal
Pig & Poultry Health Tracker Using IoT	Focuses only on IoT; misses digital management and biosecurity scope
Animal Farm Security & Management Tool	Ambiguous — 'security' is interpreted as physical security, not disease biosecurity

1.3 Project Abstract

This project proposes a web-based digital management portal designed to help pig and poultry farm owners implement, monitor, and enforce biosecurity protocols on their farms. Biosecurity — the set of practices that prevent the introduction and spread of infectious diseases — is the single most critical factor in maintaining livestock health and farm profitability.

Farms across Tamil Nadu and rural India lack structured digital tools to log biosecurity activities. Visitor entry, animal health checks, feed and medicine records, quarantine management, and vaccination schedules are typically maintained on paper or not at all. This creates critical gaps during disease outbreaks such as Avian Influenza, African Swine Fever, or Newcastle Disease.

The proposed system provides farm managers with a role-based web portal built using React.js frontend and Spring Boot backend, backed by a MySQL database. It supports real-time logging of biosecurity events, animal health tracking, visitor entry management, vaccination scheduling, incident reporting, and automated compliance alerts. An admin panel allows veterinary officers to monitor farm data across multiple registered farms.

1.4 Scope

The system covers the following within Version 1.0:

- Pig farms and poultry farms (broiler, layer, and breeder units) registered on the platform.
- Biosecurity event logging: visitor entry/exit, vehicle disinfection, foot dip usage, and perimeter checks.
- Animal health monitoring: daily mortality counts, symptom logging, and quarantine status tracking.
- Vaccination and medication schedule management with due-date alerts.
- Feed and water quality record maintenance.
- Incident reporting for disease suspicion with automatic alert to registered veterinary officer.
- Farm-level dashboard with biosecurity compliance score.
- Admin panel for veterinary officers to review farm compliance reports.

Out of Scope (Version 1.0): Market price tracking, IoT sensor integration, GPS mapping, and financial accounting are planned for Version 2.0.

1.5 Deliverable Summary — Day 1

Item	Status
Project title finalized and approved	Completed — June 01, 2025
Abstract drafted	Completed — June 01, 2025
Scope boundaries defined	Completed — June 01, 2025
Alternative titles evaluated and rejected	Completed — June 01, 2025

Day 2 | June 02, 2025 | Requirement Gathering

2.1 Problem Statement

India is one of the largest producers of poultry and pig products in Asia, with Tamil Nadu alone housing over 150 million poultry birds and a growing swine population. Despite this scale, most small and medium farms operate without any structured biosecurity framework.

Disease outbreaks such as Avian Influenza (H5N1, H5N8), Ranikhet Disease, and African Swine Fever have caused devastating losses in recent years. The primary reason these diseases spread rapidly is the absence of systematic visitor control, disinfection logging, and early symptom detection. Farm workers have no digital tool to report unusual animal behaviour, and veterinary officers have no real-time visibility into farm health status across multiple sites.

There is a critical need for a simple, role-based digital portal that a farm manager can operate on a smartphone or desktop browser to log biosecurity activities daily, receive compliance alerts, and generate reports for government veterinary inspections.

2.2 Functional Requirements

FR ID	Requirement	Description
FR-01	User Registration & Login	Farm managers, workers, and veterinary officers register with name, email, farm/institution, and role. Secure login via email+password.
FR-02	Farm Profile Management	Farm owner can register their farm with details: farm name, location (district/village), type (pig/poultry/both), total animal count.
FR-03	Visitor & Vehicle Entry Log	Farm manager logs each visitor and vehicle entering the farm: name, purpose, entry/exit time, disinfection status (yes/no).
FR-04	Daily Animal Health Check	Worker records daily mortality count, sick animal count, observed symptoms (diarrhoea, respiratory, neurological), and general flock/herd status.
FR-05	Quarantine Management	System allows creation of quarantine records for new or sick animals with start date, reason, assigned shed, and release date.
FR-06	Vaccination Schedule	Admin schedules vaccinations per species and batch. System sends in-app alerts 3 days before due date. Worker marks vaccination as completed.
FR-07	Feed & Water Quality Log	Farm manager logs daily feed quantity used, feed supplier, water source status, and any abnormalities observed.
FR-08	Incident / Disease Suspicion Report	Worker can raise an incident report with description, affected animal count, photos (optional), and urgency level. Auto-alert sent to registered veterinary officer.
FR-09	Biosecurity Compliance Score	Dashboard calculates a daily compliance score based on completed biosecurity tasks (visitor log, disinfection, health check). Score shown as percentage.
FR-10	Veterinary Officer Panel	Vet officer can view all farms assigned to them, see incident reports, download compliance reports, and add inspection notes.

2.3 Non-Functional Requirements

ID	Requirement	Detail
NFR-01	Performance	Dashboard and log submission pages must load within 3 seconds on a standard 4G mobile connection.
NFR-02	Usability	UI must function on screens as small as 360px wide. Icons and labels must be clear for workers with low digital literacy.

ID	Requirement	Detail
NFR-03	Security	Passwords stored as bcrypt hashes. JWT tokens used for session management. Role-based access enforced on all API routes.
NFR-04	Reliability	System must maintain 99% uptime during peak disease-risk seasons (monsoon: June–September, winter: November–January).
NFR-05	Scalability	Backend must handle at least 300 concurrent farm users without degradation.
NFR-06	Maintainability	All modules must be independently deployable. API must be versioned (v1) to allow future upgrades.
NFR-07	Data Integrity	All biosecurity logs must be timestamped server-side and must not be editable after 24 hours of creation to ensure audit integrity.

2.4 Stakeholder Requirements

Stakeholder	What They Need	Priority
Farm Manager / Owner	Simple daily logging interface, compliance score dashboard, vaccination alerts, incident reporting	High
Farm Worker	Easy health check form, visitor entry log, accessible on mobile phone with minimal steps	High
Veterinary Officer	View multiple farms, receive incident alerts, add inspection notes, download compliance reports	High
Project Evaluator / College	Clean code structure, documented API, ER diagram, GitHub repository with commit history	High
System Administrator	User management, farm registration approval, system health monitoring	Medium

2.5 Constraints

- The project must be completed within the academic semester timeline (June – November 2025).
- The system will be hosted on a free-tier cloud platform (Render or Railway) during evaluation phases.
- No IoT hardware integration in Version 1.0 — all data is entered manually by farm staff.
- No payment gateway or subscription billing features are included in this version.
- The system supports English language only in Version 1.0. Tamil language support is planned for Version 2.0.
- Photo uploads in incident reports are limited to 2MB per image in Version 1.0.

Day 3 | June 03, 2025 | Objective Definition

3.1 Primary Objective

To design and develop a web-based digital farm management portal that enables pig and poultry farm managers to implement, log, and monitor biosecurity measures in a structured and auditable manner, reducing the risk of disease outbreaks and enabling veterinary officers to provide timely remote oversight.

3.2 Specific Objectives

- O-1.** Design a comprehensive requirement model that captures all biosecurity workflows for pig and poultry farms through stakeholder analysis and domain research.
- O-2.** Develop a secure, role-based authentication system using JWT that supports three roles: Farm Worker, Farm Manager, and Veterinary Officer.
- O-3.** Build a Spring Boot RESTful API backend that handles biosecurity event logging, health records, vaccination scheduling, and incident reporting with full CRUD support.
- O-4.** Design and implement a normalised MySQL relational database schema that stores farm profiles, animal records, biosecurity logs, vaccination schedules, and incident reports.
- O-5.** Develop a responsive React.js frontend with dedicated pages for visitor entry, daily health check, vaccination alerts, incident reporting, and compliance dashboard — optimised for mobile use.
- O-6.** Implement an automated biosecurity compliance scoring algorithm that evaluates daily task completion and displays a farm-level score to the manager.
- O-7.** Deploy the application to a cloud platform and maintain a clean, documented GitHub repository with ER diagram, API documentation, and README that serves as the academic submission artifact.

3.3 Objectives Mapped to Deliverables

Objective	Covers	Targeted Day
O-1	Requirement gathering & stakeholder analysis	Day 1 – Day 3
O-2	Authentication & role management	Day 4 – Day 5
O-3	Spring Boot REST API development	Day 5 (CRUD APIs)
O-4	Database design, ER diagram, SQL schema	Day 6 – Day 8
O-5	React.js frontend UI design & prototype	Day 9 – Day 11
O-6	Compliance scoring module	Day 10 (Dashboard UI)
O-7	GitHub repository & documentation	Day 13 (Frontend Setup)

3.4 Expected Outcomes

- A working web portal where a farm manager can log daily biosecurity activities and see a live compliance score.
- A veterinary officer dashboard that provides real-time visibility across multiple registered farms.
- An automated alert system for vaccination due dates and incident reports that notifies the relevant officer.
- A structured MySQL database scalable to thousands of farm records and daily log entries without redesign.
- A GitHub repository with clear commit history, ER diagram, API documentation, and README suitable for academic submission.
- Documentation sufficient for another developer to set up and run the project from scratch.

Day 4 | June 04, 2025 | User & Module Identification

4.1 Identified Users

User Type 1: Farm Manager (Primary User)

The farm manager is the primary actor. They register the farm, configure settings, and are responsible for daily biosecurity logs. They view the compliance score, manage vaccination schedules, and raise incident reports. For example, a poultry farm owner in Namakkal would log morning mortality counts, record visitor entries, and check vaccination alerts each day from their smartphone.

User Type 2: Farm Worker

The farm worker has limited access. They log daily health check observations (mortality, symptoms) and record visitor entry/exit. They cannot view compliance scores, edit farm settings, or access the veterinary panel. The UI for this role is simplified — minimal screens, large buttons, Tamil-language support planned for Version 2.0.

User Type 3: Veterinary Officer

The veterinary officer accesses a read-only dashboard that aggregates data from all farms assigned to them. They receive incident alerts, can add inspection notes, and download compliance PDF reports for government records. They do not log biosecurity activities.

User Type 4: System Administrator

The administrator manages farm registration approvals, user accounts, species/vaccine master data, and system health. They do not interact with biosecurity logs directly.

4.2 System Modules

Module No.	Module Name	What It Does	Who Uses It
M-01	Authentication Module	Handles registration, login, logout, and JWT token generation for all three roles.	All Users
M-02	Farm Profile Module	Farm registration, type (pig/poultry), location, total animal capacity, and assigned vet officer.	Farm Manager, Admin
M-03	Visitor & Vehicle Log Module	Logs every person and vehicle entering the farm with disinfection and purpose details.	Farm Manager, Worker
M-04	Animal Health Check Module	Daily health record: mortality count, sick animals, observed symptoms, and shed-wise status.	Farm Worker, Manager
M-05	Quarantine Management Module	Creates and tracks quarantine records for new arrivals and sick animals.	Farm Manager
M-06	Vaccination Schedule Module	Manages vaccination due dates per batch/species, sends alerts, and records completion.	Farm Manager, Admin
M-07	Feed & Water Log Module	Tracks daily feed quantities, water source quality observations, and supplier records.	Farm Worker, Manager
M-08	Incident Reporting Module	Workers raise disease suspicion reports; auto-alerts sent to veterinary officer.	Worker, Manager, Vet
M-09	Compliance Dashboard Module	Calculates daily compliance score from completed tasks; shows farm-level performance graph.	Farm Manager, Vet
M-10	Veterinary Officer Panel	Read-only multi-farm view; incident alert inbox; compliance report download.	Veterinary Officer

Module No.	Module Name	What It Does	Who Uses It
M-11	Admin Panel Module	Farm approval, user management, master data (species, vaccines, districts), system health.	System Admin

4.3 Module Interaction Summary

When a farm worker logs in (M-01), they are directed to the simplified worker dashboard. They complete the morning health check (M-04) and log any visitors (M-03). If a serious symptom is observed, they raise an incident report (M-08), which automatically alerts the assigned veterinary officer (M-10). The farm manager reviews the compliance score on the dashboard (M-09), which is updated based on completed tasks for the day. Vaccination due alerts (M-06) appear as notifications on both the manager's and worker's screens.

Day 5 | June 05, 2025 | Use Case Diagram Preparation

5.1 Actors

- **Farm Manager** — Primary actor who registers the farm, manages biosecurity logs, views compliance score, and raises incident reports.
- **Farm Worker** — Logs daily health checks and visitor entries. Limited access role.
- **Veterinary Officer** — Receives incident alerts, views farm compliance, adds inspection notes.
- **System Administrator** — Manages users, farm registrations, and master data.
- **Notification Service** — System actor — triggers in-app alerts for vaccination due dates and incident reports.

5.2 Use Cases — Farm Manager & Worker

UC ID	Use Case Name	Description	Precondition
UC-01	Register Farm Account	Manager registers with name, email, farm details, and role. System creates account pending admin approval.	No existing account
UC-02	Login	User enters email and password. System verifies credentials and issues JWT token with role.	Account approved
UC-03	Log Visitor Entry	Manager/Worker records visitor name, purpose, entry time, and disinfection status.	Logged in
UC-04	Submit Daily Health Check	Worker records mortality count, sick animal count, and observed symptoms for the day.	Logged in
UC-05	Create Quarantine Record	Manager creates quarantine entry for new or sick animals with shed, reason, and expected release date.	Logged in as Manager
UC-06	Log Feed & Water Record	Worker logs daily feed usage, supplier, and water source quality observations.	Logged in
UC-07	Raise Incident Report	Worker raises a disease suspicion report. System auto-alerts the assigned veterinary officer.	Logged in; symptoms observed
UC-08	View Compliance Dashboard	Manager views today's biosecurity compliance score with breakdown by task category.	Logged in as Manager
UC-09	View Vaccination Schedule	Manager views upcoming vaccination due dates and marks completed vaccinations.	Logged in as Manager
UC-10	Logout	User ends the session. JWT token is invalidated.	Logged in

5.3 Use Cases — Veterinary Officer & Admin

UC ID	Use Case Name	Description
UC-11	Vet Officer Login	Officer logs in with credentials; role-verified access to veterinary panel only.
UC-12	View Farm Compliance Reports	Officer views compliance scores and trend graphs across all assigned farms.
UC-13	Receive & Respond to Incident Alert	Officer receives incident notifications, views report details, and adds inspection notes.
UC-14	Download Compliance Report PDF	Officer downloads a farm's monthly compliance report as a PDF for government records.

UC ID	Use Case Name	Description
UC-15	Admin: Approve Farm Registration	Admin reviews new farm registration requests and approves or rejects them.
UC-16	Admin: Manage Master Data	Admin adds/edits species types, vaccine names, districts, and disease categories.

5.4 CRUD API Endpoints

Auth & User APIs

Method	Endpoint	Action	Auth Required
POST	/api/auth/register	Create new user account	No
POST	/api/auth/login	Login and receive JWT token	No
GET	/api/user/profile	Read logged-in user's profile	Yes (All)
PUT	/api/user/profile	Update user profile details	Yes (All)
DELETE	/api/user/account	Delete user account	Yes (Admin)

Farm Profile APIs

Method	Endpoint	Action	Auth Required
POST	/api/farm	Register new farm	Yes (Manager)
GET	/api/farm/:id	Read farm details by ID	Yes (All)
PUT	/api/farm/:id	Update farm information	Yes (Manager)
DELETE	/api/farm/:id	Delete farm (admin only)	Yes (Admin)

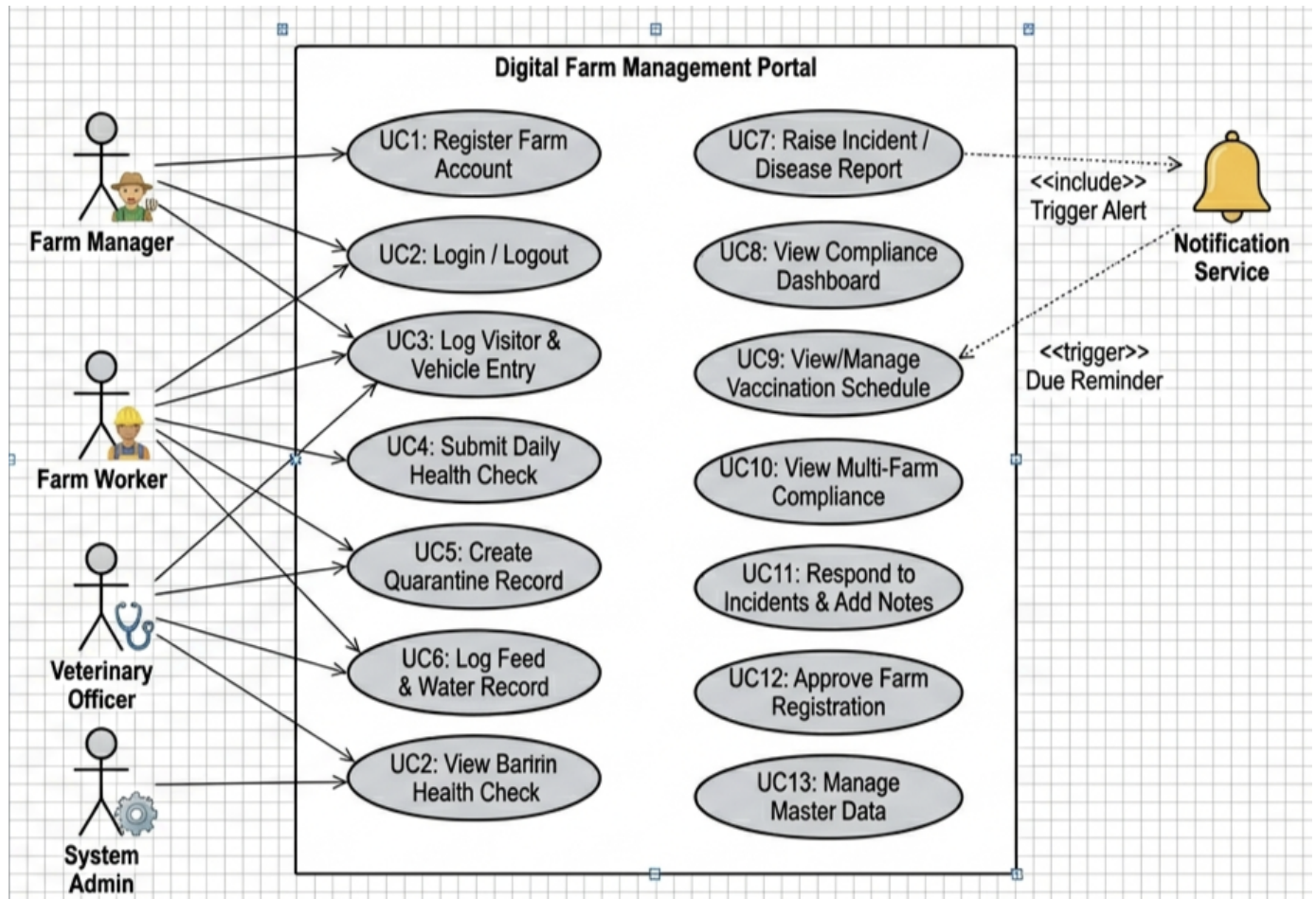
Biosecurity Log APIs

Method	Endpoint	Action	Auth Required
POST	/api/visitor-log	Create new visitor/vehicle entry log	Yes (Worker/Manager)
GET	/api/visitor-log/farm/:id	Read all visitor logs for a farm	Yes (Manager/Vet)
POST	/api/health-check	Submit daily health check record	Yes (Worker/Manager)
GET	/api/health-check/farm/:id	Read health check history for farm	Yes (Manager/Vet)
POST	/api/feed-log	Create feed and water quality log	Yes (Worker/Manager)
GET	/api/feed-log/farm/:id	Read feed log history for farm	Yes (Manager/Vet)

Incident & Vaccination APIs

5.4 Use Case Diagram

Digital Farm Management Portal — Use Case Overview



Method	Endpoint	Action	Auth Required
POST	/api/incident	Raise new incident / disease suspicion	Yes (Worker/Manager)
GET	/api/incident/:id	Read specific incident report	Yes (Manager/Vet)
PUT	/api/incident/:id	Add vet officer inspection note	Yes (Vet)
GET	/api/incident/farm/:id	Read all incidents for a farm	Yes (Manager/Vet)
POST	/api/vaccination	Create vaccination schedule entry	Yes (Manager/Admin)
GET	/api/vaccination/farm/:id	Read vaccination schedule for farm	Yes (All)
PUT	/api/vaccination/:id	Mark vaccination as completed	Yes (Manager)

Day 6 | June 06, 2025 | Database Requirement Analysis

6.1 Why We Need a Database

The Digital Farm Management Portal must persist multiple categories of data beyond a single user session. Farm profiles must be stored so managers do not need to re-register. Daily biosecurity logs — visitor entries, health checks, feed records — must be saved to build audit trails and compliance scores. Vaccination schedules must persist so alerts can be triggered on due dates. Incident reports must be stored with timestamps for veterinary review and government inspection records.

A relational database (MySQL 8.0) is chosen because the data is highly structured, the relationships between farms, animals, logs, and users are well-defined, and SQL joins are efficient for the compliance scoring and reporting features required.

6.2 Data Storage Requirements

- User and farm data must be stored securely — passwords must never be stored as plain text (bcrypt hashing required).
- Each biosecurity log (visitor, health check, feed) must be linked to the farm and the user who created it.
- Each incident report must reference the farm, the reporting user, and the assigned veterinary officer.
- Vaccination schedules must reference the species, vaccine name, batch, and farm to support due-date alerts.
- Animal records (sheds, batches) must be stored separately from health check logs to allow flexible updates.
- Compliance scores are calculated dynamically from log data; only daily scores need to be stored for trend charts.
- Quarantine records must link to animal batches and include start date, reason, and release date.
- Notifications (vaccination alerts, incident alerts) must be stored with read/unread status per user.

6.3 Identified Tables

Table Name	Stores	Key Relationships
users	All user accounts — farm managers, workers, vet officers, admins — with role flag and bcrypt password	Referenced by farms, visitor_logs, health_checks, incidents, notifications
farms	Farm profile: name, location, type (pig/poultry/both), capacity, approval status, assigned vet officer	Belongs to users (owner); referenced by all log tables
animal_batches	Individual animal groups/sheds per farm: species, batch name, entry date, current count	Belongs to farms; referenced by health_checks, quarantine_records
visitor_logs	Each visitor/vehicle entry: name, purpose, entry time, exit time, disinfection done (yes/no)	Belongs to farms and users (logged by)
health_checks	Daily animal health record: mortality count, sick count, symptoms observed, shed-wise status	Belongs to farms, animal_batches, and users
quarantine_records	Quarantine entries for new arrivals or sick animals: start date, reason, shed, release date	Belongs to farms and animal_batches
vaccination_schedules	Vaccination plan: vaccine name, species, batch, due date, completed status, completion date	Belongs to farms and animal_batches
feed_logs	Daily feed and water quality log: feed quantity, supplier, water source status, observations	Belongs to farms and users
incidents	Disease suspicion reports: description, affected count, urgency level, vet inspection notes	Belongs to farms and users (reporter and assigned vet)

Table Name	Stores	Key Relationships
compliance_scores	Daily calculated compliance score per farm: score percentage, tasks completed, tasks missed	Belongs to farms
notifications	In-app alerts: vaccination due, incident alert, inspection reminder — with read/unread status	Belongs to users

6.4 Data Volume Estimates

Table	Expected Rows (Year 1)	Growth Pattern
users	200 – 800 accounts	Steady; grows with farm registrations
farms	50 – 200 farms	Grows with platform adoption
animal_batches	200 – 1,000 batches	~3–5 batches per farm
visitor_logs	10,000 – 50,000 entries	Daily entries; ~5–10 visitors/farm/day
health_checks	18,000 – 73,000 records	One per farm per day (365 × farms)
vaccination_schedules	500 – 2,000 records	~8–10 vaccinations per batch per year
incidents	100 – 500 reports	Low volume; spikes during outbreak seasons
compliance_scores	18,000 – 73,000 records	One per farm per day
notifications	2,000 – 10,000 messages	Batch sends at vaccination due dates

Day 7 | June 07, 2025 | ER Diagram Design

7.1 Entities and Attributes

Entity: USERS

Attribute	Data Type	Constraint	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each user
full_name	VARCHAR(100)	NOT NULL	Full name of the user
email	VARCHAR(150)	NOT NULL, UNIQUE	Login email — unique across all users
password_hash	VARCHAR(255)	NOT NULL	bcrypt hashed password
role	ENUM	NOT NULL	'farm_manager', 'farm_worker', 'vet_officer', 'admin'
phone	VARCHAR(15)	NULLABLE	Mobile number for SMS alerts (future)
district	VARCHAR(100)	NULLABLE	User's district for regional filtering
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp

Entity: FARMS

Attribute	Data Type	Constraint	Description
farm_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique farm identifier
owner_id	INT	FK → users.user_id	Farm manager who registered this farm
vet_officer_id	INT	FK → users.user_id	Assigned veterinary officer
farm_name	VARCHAR(150)	NOT NULL	Name of the farm
farm_type	ENUM	NOT NULL	'pig', 'poultry', 'both'
district	VARCHAR(100)	NOT NULL	District where the farm is located
village	VARCHAR(100)	NULLABLE	Village / locality
capacity	INT	NULLABLE	Total animal capacity of the farm
is_approved	BOOLEAN	DEFAULT FALSE	Admin approval status
registered_at	TIMESTAMP	DEFAULT NOW()	Farm registration timestamp

Entity: ANIMAL_BATCHES

Attribute	Data Type	Constraint	Description
batch_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique batch identifier
farm_id	INT	FK → farms.farm_id	Farm this batch belongs to
species	VARCHAR(60)	NOT NULL	e.g., Broiler Chicken, Yorkshire Pig
batch_name	VARCHAR(100)	NOT NULL	e.g., Shed A – Batch 3

Attribute	Data Type	Constraint	Description
entry_date	DATE	NOT NULL	Date animals entered the farm/shed
current_count	INT	NOT NULL	Current live animal count in this batch
is_active	BOOLEAN	DEFAULT TRUE	Whether the batch is currently active

Entity: VISITOR_LOGS

Attribute	Data Type	Constraint	Description
log_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique log entry ID
farm_id	INT	FK → farms.farm_id	Farm where entry was recorded
logged_by	INT	FK → users.user_id	Worker/Manager who created this log
visitor_name	VARCHAR(100)	NOT NULL	Name of visitor or driver
purpose	VARCHAR(200)	NOT NULL	Purpose of visit (delivery, inspection, etc.)
entry_time	TIMESTAMP	NOT NULL	Date and time of entry
exit_time	TIMESTAMP	NULLABLE	Date and time of exit (filled later)
disinfected	BOOLEAN	NOT NULL	Whether disinfection protocol was followed

Entity: INCIDENTS

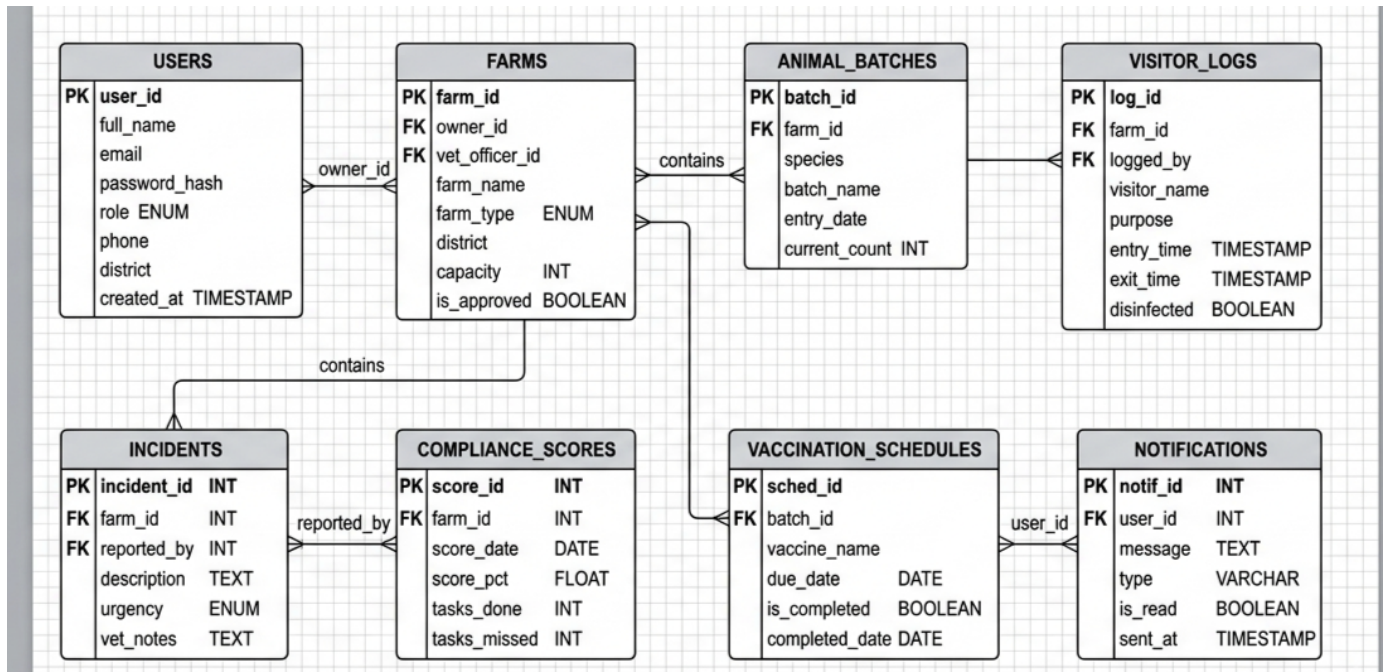
Attribute	Data Type	Constraint	Description
incident_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique incident record ID
farm_id	INT	FK → farms.farm_id	Farm where incident was observed
reported_by	INT	FK → users.user_id	Worker/Manager who raised the report
description	TEXT	NOT NULL	Description of observed symptoms or event
affected_count	INT	NULLABLE	Number of animals affected
urgency	ENUM	NOT NULL	'low', 'medium', 'high', 'critical'
vet_notes	TEXT	NULLABLE	Veterinary officer's inspection notes
reported_at	TIMESTAMP	DEFAULT NOW()	Timestamp of incident report creation

7.2 Relationships

From Entity	Relationship	To Entity	Cardinality	Meaning
USERS	Owns	FARMS	One-to-Many	One farm manager can register multiple farms; each farm has one owner.
FARMS	Contains	ANIMAL_BATCHES	One-to-Many	One farm can have many animal batches (sheds/groups).

7.3 ER Diagram

Digital Farm Management Portal — Entity Relationship Diagram



Day 8 | June 08, 2025 | Database Schema Creation

8.1 SQL Schema — Complete Table Definitions

The following SQL statements create all tables for the Digital Farm Management Portal database. Written for MySQL 8.0 compatibility.

```
-- =====
-- DATABASE: farm_biosecurity_db
-- Created:   June 08, 2025
-- Project:   Digital Farm Management Portal
-- =====

CREATE DATABASE IF NOT EXISTS farm_biosecurity_db;
USE farm_biosecurity_db;

-- TABLE 1: USERS
CREATE TABLE users (
  user_id      INT AUTO_INCREMENT PRIMARY KEY,
  full_name    VARCHAR(100) NOT NULL,
  email        VARCHAR(150) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role         ENUM('farm_manager','farm_worker','vet_officer','admin') NOT NULL,
  phone        VARCHAR(15),
  district     VARCHAR(100),
  created_at   TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- TABLE 2: FARMS
CREATE TABLE farms (
  farm_id      INT AUTO_INCREMENT PRIMARY KEY,
  owner_id     INT NOT NULL,
  vet_officer_id INT,
  farm_name    VARCHAR(150) NOT NULL,
  farm_type    ENUM('pig','poultry','both') NOT NULL,
  district     VARCHAR(100) NOT NULL,
  village      VARCHAR(100),
  capacity     INT,
  is_approved  BOOLEAN DEFAULT FALSE,
  registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (owner_id) REFERENCES users(user_id) ON DELETE CASCADE,
  FOREIGN KEY (vet_officer_id) REFERENCES users(user_id) ON DELETE SET NULL
);

-- TABLE 3: ANIMAL_BATCHES
CREATE TABLE animal_batches (
  batch_id     INT AUTO_INCREMENT PRIMARY KEY,
  farm_id      INT NOT NULL,
  species      VARCHAR(60) NOT NULL,
  batch_name   VARCHAR(100) NOT NULL,
  entry_date   DATE NOT NULL,
  current_count INT NOT NULL,
  is_active    BOOLEAN DEFAULT TRUE,
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE
);

-- TABLE 4: VISITOR_LOGS
CREATE TABLE visitor_logs (
  log_id       INT AUTO_INCREMENT PRIMARY KEY,
```

```

farm_id      INT NOT NULL,
logged_by    INT NOT NULL,
visitor_name VARCHAR(100) NOT NULL,
purpose      VARCHAR(200) NOT NULL,
entry_time   TIMESTAMP NOT NULL,
exit_time    TIMESTAMP,
disinfected  BOOLEAN NOT NULL,
FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
FOREIGN KEY (logged_by) REFERENCES users(user_id) ON DELETE CASCADE
);

-- TABLE 5: HEALTH_CHECKS
CREATE TABLE health_checks (
  check_id      INT AUTO_INCREMENT PRIMARY KEY,
  farm_id       INT NOT NULL,
  batch_id      INT,
  recorded_by   INT NOT NULL,
  check_date    DATE NOT NULL,
  mortality_count INT NOT NULL DEFAULT 0,
  sick_count    INT NOT NULL DEFAULT 0,
  symptoms      TEXT,
  general_status ENUM('normal','concern','critical') DEFAULT 'normal',
  notes         TEXT,
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
  FOREIGN KEY (batch_id) REFERENCES animal_batches(batch_id) ON DELETE SET NULL,
  FOREIGN KEY (recorded_by) REFERENCES users(user_id) ON DELETE CASCADE
);

-- TABLE 6: QUARANTINE_RECORDS
CREATE TABLE quarantine_records (
  quarantine_id INT AUTO_INCREMENT PRIMARY KEY,
  farm_id       INT NOT NULL,
  batch_id      INT NOT NULL,
  reason        VARCHAR(255) NOT NULL,
  shed_location VARCHAR(100),
  start_date    DATE NOT NULL,
  release_date  DATE,
  status        ENUM('active','released') DEFAULT 'active',
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
  FOREIGN KEY (batch_id) REFERENCES animal_batches(batch_id) ON DELETE CASCADE
);

-- TABLE 7: VACCINATION_SCHEDULES
CREATE TABLE vaccination_schedules (
  sched_id      INT AUTO_INCREMENT PRIMARY KEY,
  farm_id       INT NOT NULL,
  batch_id      INT NOT NULL,
  vaccine_name  VARCHAR(150) NOT NULL,
  species       VARCHAR(60) NOT NULL,
  due_date      DATE NOT NULL,
  is_completed  BOOLEAN DEFAULT FALSE,
  completed_date DATE,
  notes         TEXT,
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
  FOREIGN KEY (batch_id) REFERENCES animal_batches(batch_id) ON DELETE CASCADE
);

-- TABLE 8: FEED_LOGS
CREATE TABLE feed_logs (
  feed_log_id   INT AUTO_INCREMENT PRIMARY KEY,
  farm_id       INT NOT NULL,

```

```

logged_by          INT NOT NULL,
log_date           DATE NOT NULL,
feed_quantity_kg   FLOAT NOT NULL,
feed_supplier      VARCHAR(150),
water_status       ENUM('normal', 'contaminated', 'low') DEFAULT 'normal',
observations       TEXT,
FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
FOREIGN KEY (logged_by) REFERENCES users(user_id) ON DELETE CASCADE
);

```

-- TABLE 9: INCIDENTS

```

CREATE TABLE incidents (
  incident_id      INT AUTO_INCREMENT PRIMARY KEY,
  farm_id          INT NOT NULL,
  reported_by      INT NOT NULL,
  description      TEXT NOT NULL,
  affected_count   INT,
  urgency          ENUM('low', 'medium', 'high', 'critical') NOT NULL,
  vet_notes       TEXT,
  status           ENUM('open', 'under_review', 'resolved') DEFAULT 'open',
  reported_at      TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE,
  FOREIGN KEY (reported_by) REFERENCES users(user_id) ON DELETE CASCADE
);

```

-- TABLE 10: COMPLIANCE_SCORES

```

CREATE TABLE compliance_scores (
  score_id         INT AUTO_INCREMENT PRIMARY KEY,
  farm_id          INT NOT NULL,
  score_date       DATE NOT NULL,
  score_pct        FLOAT NOT NULL,
  tasks_done       INT NOT NULL DEFAULT 0,
  tasks_missed     INT NOT NULL DEFAULT 0,
  UNIQUE KEY uq_farm_date (farm_id, score_date),
  FOREIGN KEY (farm_id) REFERENCES farms(farm_id) ON DELETE CASCADE
);

```

-- TABLE 11: NOTIFICATIONS

```

CREATE TABLE notifications (
  notif_id        INT AUTO_INCREMENT PRIMARY KEY,
  user_id         INT NOT NULL,
  message         TEXT NOT NULL,
  type            VARCHAR(50) DEFAULT 'general',
  is_read         BOOLEAN DEFAULT FALSE,
  sent_at         TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);

```

9.1 Design Principles

- **Mobile-first:** All layouts designed for 360px minimum width before scaling up to desktop.
- **One action per screen:** Each page has one primary goal. The visitor log page does nothing except capture visitor details.
- **Plain language:** No technical jargon in labels. Use 'Did the visitor disinfect?' not 'Disinfection protocol status (boolean)'.
- **Colour theme:** Green tones (agriculture, safety, health). Status indicators: green = normal, amber = concern, red = critical.
- **Accessibility:** Large touch targets (minimum 44px), high contrast text, and icon+label pairs for low-literacy workers.

9.2 Page Layout Wireframes

Page 1: Login Screen

[illegible]

Page 2: Daily Health Check Form

← Daily Health Check	
Farm: Rajan Poultry Farm, Namakkal	
Date: June 09, 2025	
Select Batch/Shed: [Shed A - Broiler]	
Morning Mortality Count	[_____]
Sick Animals Count	[_____]
Observed Symptoms:	
☐ Respiratory	☐ Diarrhoea
☐ Neurological	☐ Reduced Feed
☐ Sudden Death	☐ Skin Lesions
General Status:	
☐ Normal	☐ Concern ☐ Critical
Additional Notes: [_____]	
[SUBMIT HEALTH CHECK] (green btn)	

Page 3: Visitor Entry Log

[←] Log Visitor Entry	
Visitor / Driver Name	[_____]
Purpose of Visit	[_____]
Vehicle Number	[_____]
Entry Time	[09:30 AM]
Disinfection Done?	
() Yes – Foot dip + spray completed	
() No – Reason: [_____]	
[SAVE ENTRY LOG] (green button)	
Today's Entries (3)	
08:15 Feed Delivery ✓ Dis	
09:00 Vet Visit ✓ Dis	

Page 4: Incident Report Screen

[←] Report Disease Suspicion	
Description of Observation:	
(multi-line text area)	
Number of Animals Affected: [_____]	
Urgency Level:	
() Low () Medium (●) High	
[+ Add Photo (optional)]	
This report will immediately alert your assigned veterinary officer.	
[SUBMIT INCIDENT REPORT]	
(red warning-style button)	

10.1 Login Screen — Component Specification

Component	Type	Validation	Behaviour
Email field	Text input	Must match email format	Shows red border + error text below if invalid
Password field	Password input	Min 8 characters	Eye icon toggle; strength indicator shown on register page
Login button	Submit button	Both fields filled	Disabled and shows spinner during API call
Register link	Navigation link	None	Navigates to /register page
Error message	Alert box	None	Appears below button if credentials are wrong; auto-dismissed after 5s

10.2 Farm Manager Dashboard — Screen Layout

[Leaf Logo] Farm Biosecurity Portal [Bell3] [Profile]			
Good morning, Rajan! Namakkal, Tamil Nadu			
Today's Score		Open Incidents	
87%		2	
Vaccinations Due		Total Batches	
3 this week		5 Active	
[+ LOG VISITOR ENTRY] [+ HEALTH CHECK]			
Recent Biosecurity Activity			
Jun-09 Health Check Shed A Status: Normal			
Jun-09 Visitor Log 3 entriesDisinfected: All			
Jun-08 Vaccination NewcastleCompleted			
[View Full Activity Log]			

10.3 Dashboard Component Breakdown

Component	Purpose	Data Source
Top navigation bar	Shows portal logo, notification bell (with unread count), and profile dropdown	Static + auth context + notifications table (COUNT unread)
Greeting with location	Personalises the page using manager's name and farm district	users table + farms table
Today's Compliance Score card	Shows today's biosecurity compliance score as a percentage	compliance_scores WHERE farm_id AND score_date = TODAY()

Component	Purpose	Data Source
Open Incidents card	Count of unresolved incident reports for this farm	incidents WHERE farm_id AND status = 'open'
Vaccinations Due card	Number of vaccinations due within the next 7 days	vaccination_schedules WHERE due_date <= DATE_ADD(NOW(), INTERVAL 7 DAY)
Quick Action buttons	Primary CTAs: Log Visitor Entry and Health Check	Navigation links to /visitor-log and /health-check
Recent Activity list	Shows last 5 biosecurity events across all log types	UNION query across visitor_logs, health_checks, vaccination_schedules (LIMIT 5)

11.1 Navigation Structure

Route	Page	Access
/	Landing / Login redirect	Public
/login	User login page	Public (redirects to /dashboard if logged in)
/register	Farm manager registration	Public
/dashboard	Farm manager home screen	Protected (Manager role)
/visitor-log	Visitor & vehicle entry form	Protected (Manager / Worker)
/health-check	Daily animal health check form	Protected (Manager / Worker)
/health-check/history	Health check log history	Protected (Manager)
/quarantine	Quarantine record management	Protected (Manager)
/vaccination	Vaccination schedule list and form	Protected (Manager)
/feed-log	Feed and water quality log form	Protected (Manager / Worker)
/incident/new	New incident / disease suspicion form	Protected (Manager / Worker)
/incident/history	Incident report history list	Protected (Manager / Vet)
/compliance	Compliance score trend graph	Protected (Manager)
/vet/login	Veterinary officer login page	Public (vet role only)
/vet/dashboard	Multi-farm compliance overview	Protected (Vet Officer)
/vet/incidents	Incident alert inbox for vet officer	Protected (Vet Officer)
/admin/dashboard	Admin panel home	Protected (Admin role)
/admin/farms	Farm registration approvals	Protected (Admin)
/admin/users	User management	Protected (Admin)

11.2 Form Validation Rules — Health Check Form

Field	Rule	Error Message Shown to User
Batch / Shed	Must select from dropdown	Please select a batch or shed before submitting
Mortality Count	Min: 0, Max: current_count	Mortality count cannot exceed total animals in this batch
Sick Animals Count	Min: 0, Max: current_count	Sick count cannot exceed total animals in this batch
Symptoms (checkboxes)	At least one if sick_count > 0	Please select at least one observed symptom if sick animals are reported
General Status	Required radio selection	Please select the general health status for this batch today
Check Date	Cannot be future date	Health check date cannot be in the future

11.3 Prototype Flow

1. Farm manager opens the app → lands on Login page.
2. After login → redirected to Dashboard with today's compliance score and recent activity.
3. Manager taps '+ Health Check' → navigates to Daily Health Check form.
4. Manager selects the batch (Shed A – Broiler), enters mortality count (2), sick count (5), checks 'Respiratory' symptom, sets status to 'Concern'.
5. If any value fails validation, the field turns red with the error message shown inline below the field.
6. Manager taps 'Submit Health Check'. A loading spinner appears during the API call.
7. On success, manager is returned to the Dashboard. The compliance score updates to reflect the completed health check task.
8. If status is 'Critical', system automatically prompts manager to raise an Incident Report.
9. Manager taps '+ Log Visitor Entry' → fills visitor name, purpose, disinfection status → saves.
10. All completed tasks for the day are reflected in the Compliance Score (%) on the dashboard.

Day 12 | June 12, 2025 | Design Review

12.1 Review Summary

All design artifacts created from Day 1 to Day 11 were reviewed on June 12, 2025. The review verified that each deliverable is internally consistent, complete, and ready for the development phase beginning on June 14, 2025.

12.2 Review Checklist

Day	Deliverable	Review Finding	Status
1	Approved Project Title	Title is domain-specific, academically appropriate, and clearly states biosecurity scope	Approved
2	Problem Statement	Problem is grounded in real Indian pig and poultry farming challenges; not generic	Approved
3	Project Objectives	7 SMART objectives mapped to deliverables; compliance scoring objective explicitly included	Approved
4	Module List	11 modules identified; each has clear owner, purpose, and user role mapping	Approved
5	Use Cases & CRUD APIs	All major biosecurity actions covered; APIs follow REST conventions and role-based auth	Approved
6	Database Table List	All 11 tables justified; volume estimates realistic for Year 1 farm deployment	Approved
7	ER Diagram	All foreign key relationships correct; no many-to-many without junction table	Approved
8	SQL Schema	All FK constraints present; ON DELETE behaviour specified for all relationships	Approved
9	UI Wireframes	4 key wireframes complete; design principles documented; mobile-first confirmed	Approved
10	Login & Dashboard UI	All components specified; compliance score data source query written	Approved
11	Navigation & Form Design	All 19 routes listed; health check form validation rules complete	Approved

12.3 Issues Found and Resolved

- During ER diagram review, it was found that the original design did not include an ON DELETE SET NULL rule for the `vet_officer_id` foreign key in the `farms` table. This was corrected so that deleting a vet officer account does not cascade-delete the farm record.
- The health check form wireframe initially did not include a batch/shed selector, making it ambiguous which animals the record applied to. A dropdown for batch selection was added as a required field.
- The admin panel was originally designed to share the same users table with a simple role column. On review, this was confirmed as correct — role-based JWT enforcement is sufficient to prevent privilege escalation without a separate table.
- The compliance score calculation logic was not defined in Day 9–11. During review, it was clarified: $\text{score} = (\text{tasks_completed} / \text{total_daily_tasks}) \times 100$, where daily tasks are: health check submitted, visitor log updated, and feed log submitted.

12.4 Sign-Off

Document Section	Approved By	Date
Project Title and Abstract	Project Developer	June 12, 2025
Requirements and Objectives	Project Developer	June 12, 2025
ER Diagram and SQL Schema	Project Developer	June 12, 2025
UI Wireframes and Prototype Flow	Project Developer	June 12, 2025

Day 13 | June 13, 2025 | Frontend Environment Setup

13.1 Setup Steps Completed Today

Step 1. Initialised the React project using Vite (chosen over Create React App for faster build times and better HMR performance):

```
npm create vite@latest farm-biosecurity-portal -- --template react
cd farm-biosecurity-portal
npm install
```

Step 2. Installed and configured Tailwind CSS for utility-first styling:

```
npm install -D tailwindcss postcss autoprefixer
npx tailwindcss init -p
```

Step 3. Installed React Router DOM (v6) for client-side routing and Axios for HTTP API calls:

```
npm install react-router-dom axios
```

Step 4. Created full project folder structure as planned on Day 11.

Step 5. Created .env file with VITE_API_BASE_URL variable pointing to the Spring Boot backend.

Step 6. Initialised Git repository, created .gitignore (excluded node_modules, .env, dist/), made first commit.

Step 7. Created GitHub repository named 'digital-farm-biosecurity-portal', pushed initial commit with README.

13.2 Folder Structure Created

```
farm-biosecurity-portal/
├── public/
│   └── vite.svg
├── src/
│   ├── components/
│   │   ├── Navbar.jsx           ← Top navigation bar with notification bell
│   │   ├── VisitorLogForm.jsx   ← Visitor & vehicle entry form component
│   │   ├── HealthCheckForm.jsx  ← Daily animal health check form
│   │   ├── IncidentForm.jsx     ← Disease suspicion / incident report form
│   │   ├── ComplianceCard.jsx   ← Today's compliance score card
│   │   └── ActivityItem.jsx     ← Single row in recent activity list
│   ├── pages/
│   │   ├── Login.jsx
│   │   ├── Register.jsx
│   │   ├── Dashboard.jsx
│   │   ├── VisitorLog.jsx
│   │   ├── HealthCheck.jsx
│   │   ├── HealthCheckHistory.jsx
│   │   ├── Quarantine.jsx
│   │   ├── Vaccination.jsx
│   │   ├── FeedLog.jsx
│   │   ├── IncidentNew.jsx
│   │   ├── IncidentHistory.jsx
│   │   ├── Compliance.jsx
│   │   └── vet/
│   │       ├── VetDashboard.jsx
│   │       └── VetIncidents.jsx
│   ├── services/
│   │   └── api.js               ← All Axios API call functions
│   ├── context/
│   │   └── AuthContext.jsx      ← JWT token, user role, farm context
│   ├── hooks/
│   │   └── useAuth.js           ← Custom auth hook for protected routes
```

- ■■■ assets/

■ ■■■ App.jsx

■ ■■■ main.jsx

■■■ .env

■■■ .gitignore

■■■ package.json

■■■ tailwind.config.js

■■■ README.md
- ← Route definitions (React Router v6)

← Entry point

← VITE_API_BASE_URL=http://localhost:8080

13.3 Sprint Completion Summary

Phase	Days	Status
Project Definition (Title, Requirements, Objectives)	Day 1 – Day 3	Complete
System Design (Modules, Use Cases, APIs)	Day 4 – Day 5	Complete
Database Design (Table List, ER Diagram, SQL Schema)	Day 6 – Day 8	Complete
UI Design (Wireframes, Screens, Prototype)	Day 9 – Day 11	Complete
Review and Environment Setup	Day 12 – Day 13	Complete